

OX-MABSR: Appendix

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The supplementary material provides all the prompts used in this work, along with additional experimental details. Specifically, it includes:

- **Data Construction Prompt:** This section provides the prompts used to guide annotation generation and annotation verification.
- **Semantic Coherence Score Prompt:** This section details the prompt template employed to compute the semantic coherence score for model-generated two-level sentiment explanations and reasoning paths.
- **Task Prompts:** This section includes the input prompts designed for various tasks in the OX-MABSR.
- **Evaluation on MABSA:** This section presents additional results of our model on the Multimodal Aspect-Based Sentiment Analysis to demonstrate effectiveness.
- **More Case Studies:** This section showcases further qualitative examples to illustrate the reasoning capability and comparative performance of different models.
- **Ethic Considerations:** This section elaborates on ethical practices in data handling, fairness, and bias mitigation throughout benchmark construction and model development.

A Data Construction Prompt

A.1 Annotation Generation Prompt

To provide clarity on our dataset construction pipeline, we present below the detailed prompt designs used to guide the MLLM during annotation generation. Our annotation generation consists of five steps, each targeting distinct elements: Event Extraction, Aspect detection, Perceptual-level Reasoning, Cognitive-level Reasoning, and Sentiments Generation. The following sections describe the objective of each step and present the corresponding prompt used during annotation generation.

Step 1: Event Extraction. To extract salient events from each image-text pair, we prompt the MLLM to identify the main happenings or situations described in the multimodal input. These events serve as narrative anchors that ground the downstream sentiment reasoning in a specific situational context. The extracted events are typically textual descriptions capturing the core actions, interactions, or happenings depicted or implied.

►Input:

Image: {image}, Sentence: {text}

►Prompt:

Give you an image and a sentence, you can provide historical context, important events, and relevant background information related to the image and the sentence. Summarize the historical context, important events, and relevant background text accurately and objectively. Your summary should: 1 Be between 100 and 150 words; 2 Avoid subjective interpretation or modern bias; 3 Maintain factual accuracy and chronological clarity.

►Output:

Event: {event}

Step 2: Aspect Detection. Based on the previously extracted events, we prompt the MLLM to detect relevant aspects that may carry sentiment. These aspects can be explicit or implicit, and are expected to be the sentiment-bearing aspects or concepts within the event. The identification of aspects ensures that the sentiment reasoning is anchored to meaningful and interpretable targets.

►Input:

Image: {image}, Sentence: {text}, Event: {event}

►Prompt:

Please extract all sentiment-bearing aspects from the following text. 1 An ‘aspect’ can be a person, product, brand, location, or event. 2 Only include aspects that are associated with a clear sentiment. 3 Output the results in the following format: [aspect1, aspect2, aspect3...].

►Output:

Aspects: {[aspect1, aspect2, aspect3...]}

Step 3: Perceptual-level Reasoning. We guide the MLLM to perform surface-level analysis of observable features, including visual aesthetics, facial expressions, scene layout, and text semantics. For each aspect, the model is prompted to identify intuitive perceptual causes that may explain the expressed or perceived sentiment. These causes reflect immediate impressions and visually salient cues.

►Input:

Image: {image}, Sentence: {text}

►Prompt:

Please provide a detailed and aesthetic description of the given image. Focus on visual elements such as lighting, color tone, composition, textures, spatial depth, and emotional atmosphere. Describe how these elements contribute to the overall aesthetic and emotional impact of the image. For example: "The lighting of the image is very [MASK]. The colors are [MASK], creating a [MASK] mood. The composition feels [MASK], with elements arranged in a [MASK] manner. The textures appear [MASK], adding a sense of [MASK]. There is a sense of depth created by [MASK]. Overall, the image evokes a feeling of [MASK], with a visual style that can be described as [MASK]." Use specific, vivid adjectives and avoid generic or vague descriptions.

►Output:

Aesthetic Information: {aesthetic information}

►Input:

Image: {image}, Sentence: {text}, Aspects: {[aspect1, aspect2, aspect3...]}, Aesthetic Information: {aesthetic information}

►Prompt:

Given an image and a sentence, please use the aesthetic information of the image, the facial expressions of the characters in the image, the scene information of the image, and the textual semantics. Your task is to determine each aspect's sentiment and extract the cause for that sentiment based on the context provided. For each aspect, please output one sentence using the following format: "aspect's sentiment is [SENTIMENT], and the cause for this sentiment is [CAUSE]." Make sure to: 1 Use the provided Aspects without modification. 2 Sentiments can be complex, consisting of one or more sentiment labels. 3 Clearly state the causes for the sentiment based on the information given in the text. 4 Generate a separate sentence for each aspect-sentiment pair. 5 Incorporate both visual and textual clues to justify your reasoning, even if the cause is implicit. 6 Ensure your response is coherent, contextually grounded, and avoids generic or vague explanations.

►Output:

Perceptual Causes: {perceptual causes}

Step 4: Cognitive-level Reasoning. In this step, the MLLM is instructed to go beyond surface cues and explore deeper contextual knowledge. It is prompted to infer background causes like prior events, social context, or implicit references, providing abstract, cognitive-level explanations that complement perceptual ones.

►Input:

Image: {image}, Sentence: {text}, Aspects: {[aspect1, aspect2, aspect3...]}, Event: {event}

►Prompt:

Given an image and a sentence, please use the historical context, important events, and relevant background information of the image. Your task is to determine each aspect's sentiment and extract the cause for that sentiment based on the context provided. For each aspect, please output one sentence using the following format: "aspect's sentiment is [SENTIMENT], and the causes for this sentiment are [CAUSE]." Make sure to: 1 Use the provided aspects. 2 Sentiments can be complex, consisting of one or more sentiment labels. 3 Clearly state the cause for the sentiment based on the information given in the text. 4 Generate a separate sentence for each aspect-sentiment pair. 5 Integrate implicit knowledge or commonsense reasoning if necessary to explain deeper motivations behind the sentiment.

►Output:

Cognitive Causes: {cognitive causes}

Step 5: Sentiments Generation. The MLLM then integrates perceptual and cognitive causes to generate open-vocabulary sentiment labels for each aspect, producing expressive sentiment that reflect its overall understanding.

►Input:

Image: {image}, Sentence: {text}, Aspects: {[aspect1, aspect2, aspect3...]}, Perceptual Causes: {perceptual causes}, Cognitive Causes: {cognitive causes}

►Prompt:

Given an image, a sentence, and a list of aspects, your task is to generate sentiment annotations for each aspect. Please integrate both perceptual-level causes and cognitive-level causes to infer complex and expressive sentiments. Instead of categorizing sentiment as simply Positive, Negative, or Neutral, use open-vocabulary, fine-grained sentiment words (e.g., grief, admiration, relief, shame). Each aspect can correspond to one or more sentiments, reflecting sentiment complexity. Please output your response in the following structured format: [[aspect1, (sentiment1, sentiment2, ...)], [aspect2, (sentiment1, sentiment2, ...)], ...], 1 Use the provided aspects exactly as given. 2 For each aspect, output a tuple of nuanced sentiment words. 3 All sentiment words should be expressive, specific, and human-like. 4 Do not include causes in the output, only aspect-sentiment pairs in the required format. 5 Ensure that sentiments are grounded in both visual and textual context.

►Output:

Sentiments: {[[aspect1, (sentiment1, sentiment2, ...)], [aspect2, (sentiment1, sentiment2, ...)], ...]}

A.2 Annotation Verification Prompt

We present below the detailed prompt designs used to guide the MLLM during annotation verification. Our annotation verification consists of three steps, each designed to engage the MLLM in a distinct mode of reasoning. The following sections describe the objective of each step and present the corresponding prompt used during annotation verification.

Step 1: Supportive Reasoning. We guide the MLLM to assume that a given annotation is correct and to generate supportive reasons for it. The model is prompted to retrieve contextual and perceptual evidence from both the image and the text, highlighting observations that justify the annotation. This step elicits intuitive and interpretable reasons that align with the annotation, serving as an internal check for plausibility.

►**Input:**

Image: {image}, Sentence: {text},
Annotations: {Aspects, Sentiments, Perceptual Causes, Cognitive Causes}

►**Prompt:**

You are given an image, a sentence, and an aspect sentiment annotation along with its explanation. Your task is to assume that the annotation is correct and explain why this annotation can be justified based on the given multimodal context. In your response, please provide detailed, step-by-step reasoning that draws on both the image and the sentence. Consider the following in your explanation:

- 1 Visual cues: How do visual elements (e.g., facial expressions, body language, scene context, object interactions, composition) support the annotation?
- 2 Textual cues: How do specific words, phrases, or implications in the sentence support the annotation toward the entity or aspect?
- 3 Cross-modal alignment: How do the image and text work together to reinforce the annotation or clarify the emotional context?
- 4 Consistency and plausibility: Why is the annotation logically and contextually consistent with the input?
- 5 Be clear and detailed, providing evidence that supports the annotation and shows how the annotation fits the context.

►**Output:**

Supportive Reasons: {supportive reasons}

Step 2: Critical Reasoning. In this step, the MLLM is instructed to assume the annotation may be incorrect and to critically examine it. The model is prompted to identify inconsistencies, weak justifications, or contradictory evidence within the image-text context. This encourages the model to generate opposite reasons, helping to uncover potential hallucinations or over-interpretations in the annotation.

►**Input:**

Image: {image}, Sentence: {text}, Annotations: {Aspects, Sentiments, Perceptual Causes, Cognitive Causes}

►**Prompt:**

You are given an image, a sentence, and an aspect sentiment annotation along with its explanation. Your task is to assume that the annotation is incorrect and explain why this annotation cannot be justified based on the provided multimodal context. In your explanation, please provide a step-by-step analysis that examines both the image and the sentence critically. Consider the following points: 1 Visual contradictions: Identify any facial expressions, body language, scene elements, visual details that contradict the annotation. 2 Textual inconsistencies: Identify words or cues that contradict the given annotation. 3 Cross-modal conflicts: Explain how the image-text pair exposes issues with the sentiment annotation. 4 Potential hallucinations: Point out if the annotation relies on unsupported assumptions.

►**Output:**

Opposite Reasons: {opposite reasons}

Step 3: Reflective Judgment. The MLLM reflects on both supportive and opposite reasons to decide whether to accept or reject the original annotation.

►**Input:**

Image: {image}, Sentence: {text}, Annotations: {Aspects, Sentiments, Perceptual Causes, Cognitive Causes}, Supportive Reasons: {supportive reasons}, Opposite Reasons: {opposite reasons}

►**Prompt:**

You are given an image, a sentence, and an aspect sentiment annotation along with its explanation. Your task is to reflect on the annotation by evaluating both supportive and opposite reasons. Carefully consider how the image and sentence interact to support or contradict the sentiment and its explanation. In your response, follow these steps: 1 Analyze supportive reasons: Summarize and assess whether the supporting reasons are well-grounded in the image-text context. 2 Analyze opposite reasons: Summarize and assess whether the opposite reasons reveals valid concerns or conflicting evidence. 3 Compare both sides: Weigh both arguments and judge which is more compelling given the multimodal context. 4 Make a final decision: Decide whether to accept or reject the annotation based on your analysis. 5 Be logical, balanced, and evidence-driven in your judgment. Your final sentence must clearly state the decision in the following format: Final decision: Accept or Final decision: Reject.

►**Output:**

Final Decision: {Accept or Reject}

A.3 Reasoning Data Construction Predefined Template

The predefined template serves as a structured scaffold to organize intermediate outputs into a coherent reasoning paths. It highlights distinct layers of reasoning, including perceptual evidence (e.g., visual or textual cues), contextual cues (e.g., dialogue flow or situational details), and cognitive inferences (e.g., inferred motivations or social norms). By enforcing consistency in format and content, the template ensures that the resulting reasoning paths are interpretable, and suitable for downstream training or evaluation. This structured format enables fine-grained analysis of how each step contributes to the final sentiment judgment.

“<think>Let’s solve this problem step by step. Firstly, we need to obtain the historical background information related to the image and sentence: ”+ **Event**+ “Secondly, we extract from the historical background information the relevant aspects: ”+ **Aspects**+ “Thirdly, we obtain the surface information related to the image and sentence, that is, the intuitive visual experience of the image including the aesthetic information of the image:”+ **Aesthetic Information** + “, the facial expressions of the characters, the scene information of the image, and the intuitive meaning of the text. Next, we identify each aspect’s sentiment and the intuitive causes behind it:”+ **Perceptual Causes** + “Finally, we utilize the historical background information from the previous step to analyze each aspect’s sentiment and their deeper perspective causes:”+ **Cognitive Causes** + “Therefore, the final aspects and sentiment that can be obtained are:</think>”+ **Aspects-Sentiments Pairs**.

B Semantic Coherence Score Prompt

To evaluate the semantic coherence between the model’s prediction and the ground-truth output, we employ a prompt-based evaluation strategy using Qwen3. Instead of relying solely on token-level overlap or surface-form similarity, our approach leverages LLM to perform nuanced, structure-aware judgment. The full prompt template is shown below:

►**Input:**
Gold: {True Output}, Prediction: {Predicted Output}
►**Prompt:**
Given the following two perceptual sentiment causes, score how semantically consistent, logically coherent, and sentiment aligned the prediction is compared to the gold. Gold: {Gold}, Prediction: {Prediction}. Return only a number between 0 and 1 (inclusive). Do not include any explanation or extra text. Just output the number. Score:
►**Output:**
Perceptual Score: {Semantic Coherence Score}

►**Input:**

Gold: {True Output}, Prediction: {Predicted Output}

►**Prompt:**

Given the following two cognitive sentiment causes, score how semantically consistent, logically coherent, and sentiment aligned the prediction is compared to the gold. Gold: {Gold}, Prediction: {Prediction}. Return only a number between 0 and 1 (inclusive). Do not include any explanation or extra text. Just output the number. Score:

►**Output:**

Cognitive Score: {Semantic Coherence Score}

►**Input:**

Gold: {True Output}, Prediction: {Predicted Output}

►**Prompt:**

Given the following two reasoning paths, score how semantically consistent, logically coherent, and sentiment aligned the prediction is compared to the gold. Gold: {Gold}, Prediction: {Prediction}. Return only a number between 0 and 1 (inclusive). Do not include any explanation or extra text. Just output the number. Score:

►**Output:**

Reasoning Paths Score: {Semantic Coherence Score}

C Task Prompts

To ensure a fair and consistent comparison across different MLLMs, we construct a unified prompting framework for all tasks in OX-MABSR. Each model is provided with the same multimodal inputs and task prompts. The prompt is designed to guide models through complex sentiment reasoning over both visual and textual modalities, encouraging structured inference and minimizing ambiguity. The full prompt used is shown below:

Task 1: Open-vocabulary aspect-sentiment pairs prediction.

►**Input:**

Image: {image}, Sentence: {text}

►**Prompt:**

Extract Explicit and Implicit aspects and their Complex Sentiment: Given an image and a sentence, please identify the aspects and sentiment associated with this image and sentence. Output the results in the following format: [(aspect1, complex sentiment1, complex sentiment2)].

►**Output:**

Aspect-Sentiment Pairs: {[(aspect1, complex sentiment1, complex sentiment2...), (aspect2, complex sentiment1, complex sentiment2...)...]}

Task 2: Two-level sentiment explanations generation task.

►Input:

Image: {image}, Sentence: {text}

►Prompt:

Shallow reasoning: Given an image and a sentence, please judge the aspect and its sentiment, and explain the surface causes for this sentiment.

►Output:

Perceptual Causes: {Causes}

►Input:

Image: {image}, Sentence: {text}

►Prompt:

Deep reasoning: Given an image and a sentence, please judge the aspect and its sentiment, and explain the deep causes for this sentiment.

►Output:

Cognitive Causes: {Causes}

Task 3: The reasoning paths generation task.

►Input:

Image: {image}, Sentence: {text}

►Prompt:

Given an image and a sentence, please think step by step, obtain the sentiments and causes of the aspects from multiple levels and angles, and finally output the list of aspects and sentiments.

►Output:

Reasoning Paths: {Paths}

D Evaluation on MABSA

To further evaluate the generalization ability of our model, we test MABSR-LLM on the Multimodal Aspect-Based Sentiment Analysis (MABSA) task. This task requires the model to identify fine-grained aspect terms in a sentence and determine their corresponding sentiment polarity (positive, neutral, or negative), while also taking into account visual context.

Although MABSR-LLM is primarily trained for OX-MABSR, its step-wise reasoning and multimodal understanding enable it to adapt well to MABSA. We evaluate the model on two widely used datasets: Twitter-15 and Twitter-17 (Yu and Jiang 2019), both of which contain image-text pairs annotated with aspect terms and their associated sentiment labels.

During inference, the model is provided with both the image and text, as well as a task prompt, enabling it to align visual and textual clues while performing fine-grained sentiment analysis. The full prompt is shown below:

Method	TWITTER-15			TWITTER-17		
	P	R	F1	P	R	F1
VLP (Ling and Yu 2022)	65.1	68.3	66.6	66.9	69.2	68.0
AoM (Zhou et al. 2023)	67.9	69.3	68.6	68.4	71.0	69.7
M2DF (Zhao et al. 2023)	67.0	67.3	67.6	67.9	68.8	68.3
Atlantis (Xiao et al. 2024)	65.6	69.2	67.3	68.6	70.3	69.4
DQPSA (Peng et al. 2024)	71.7	72.0	71.9	71.1	70.2	70.6
AESAL (Zhu et al. 2024)	68.7	70.4	69.5	69.4	74.8	72.0
CORSA (Liu et al. 2025)	69.0	70.8	69.9	70.1	71.0	70.6
DaNet (Zhu et al. 2025a)	70.8	71.5	71.2	71.3	72.9	72.1
DEQA (Han et al. 2025)	71.4	73.9	72.7	71.4	72.4	71.9
AETS (Zhu et al. 2025b)	69.7	74.7	72.1	72.6	73.7	73.1
MABSR-LLM (Ours)	79.3	80.9	80.1	79.1	81.4	80.2

Table 1: Performance comparison on MABSA task.

►Input:

Image: {image}, Sentence: {text}

►Prompt:

Extract the Explicit aspect and its Simple Sentiment(POS or NEU or NEG): Given an image and a sentence, please identify the aspect and sentiment in the sentence. Output the results in the following format: [[aspect1, sentiment], [aspect2, sentiment],...].

►Output:

Aspect-Sentiment Pairs: {[[aspect1, sentiment], [aspect2, sentiment],...]}

We compare MABSR-LLM to several strong baselines, including recent state-of-the-art models such as AETS (Zhu et al. 2025b), DEQA (Han et al. 2025), and DaNet (Zhu et al. 2025a). As shown in Table 1, MABSR-LLM achieves new state-of-the-art results on both datasets. On Twitter-15, our model achieves an F1-score of 80.1, outperforming the previous best model (DEQA, 72.7) by 7.4 F1. On Twitter-17, our model obtains an F1-score of 80.2, exceeding the best baseline (AETS, 73.1) by 7.1 F1.

Our model is specifically trained for the newly proposed OX-MABSR task, which significantly extends the traditional MABSA setting. It is designed to learn (1) open-vocabulary aspect-sentiment pairs prediction, (2) the generation of both perceptual and cognitive causes of sentiment, and (3) the reconstruction of sentiment formation trajectories through reasoning paths. These tasks endow our model with a strong capacity for sentiment understanding. As a result, it achieves superior performance on standard MABSA tasks, which are essentially simplified cases of our proposed framework.

E More Case Studies

To further assess the sentiment reasoning capability of our model, we conduct additional case studies on the OX-MABSR task. We select representative image-text cases that cover open-vocabulary aspect-sentiment and sentiment explanations, allowing us to analyze the strengths of MABSR-LLM in diverse settings.

Specifically, we compare our MABSR-LLM with the

Image		Text		
Input		#Promise #focus on the #road @LetsGoToTheEX @CAA #Toronto #2015 #Drivesafe #Ontario #canada #CNE2015		
Ground Truth		MABSR-LLM (Ours)	InternVL3-8B	Qwen2.5-VL-7B
Aspect-Sentiment	[('Canadian Automobile Association', ['Supportive', 'Encouraging']), ('Workplace Safety North', ['Supportive', 'Uplifting'])]	[('Canadian Automobile Association', ['Supportive', 'Encouraging']), ('Workplace Safety North', ['Supportive', 'Uplifting'])]	(Promise Focus on the Road, Positive)	(#road @LetsGoToTheEX, Positive)
Perceptual Causes	Canadian Automobile Association's sentiment is positive, supportive, and responsible, caused by its promotion of road safety through the campaign "Promise to Focus on the Road" which aims to reduce driving distractions and foster a culture of safe driving. Workplace Safety North's sentiment is professional, uplifting, and committed, caused by its involvement in educating drivers and employers about workplace-related road safety, contributing to an engaging approach to raising awareness.	Canadian Automobile Association's sentiment is positive, supportive, and encouraging, caused by its active promotion of road safety through initiatives like "Promise to Focus on the Road," as well as its visible presence at the event with a professional and approachable image. Workplace Safety North's sentiment is professional, uplifting, and committed, and the cause for this sentiment is the emphasis on driver distraction reduction and the use of hashtags like "#Drivesafe," which reflect a dedication to ensuring safety in work-related travel contexts.	The image shows a group of people holding signs with positive messages related to road safety and promises. The hashtags and entities mentioned in the image, such as CAA, Ontario, Canada. The overall theme of the image is promoting road safety and making a promise to drive safely, which is a positive sentiment.	- **[#Promise #focus]**: The use of "promise" and "focus" implies a commitment to safety and attention, which are positive sentiments. - **[#road @LetsGoToTheEX]**: Refers to road safety and promoting travel to the EX (Exhibition), combining road safety and positivity for attending an event.- Other hashtags like **[#Toronto #2015]** and **[#CNE2015]** provide context but do not carry inherent sentiment. The image promotes safe driving habits through promises of focus, likely at a road safety event, enhancing community awareness positively.
Cognitive Causes	Canadian Automobile Association's sentiment is Encouraging and Professional, caused by its commitment to public welfare and safety through promoting road safety and educating drivers about safe practices. Workplace Safety North's sentiment is Supportive and Professional, caused by its professional dedication to reducing accidents and promoting safe behaviors as part of its duty and responsibility in workplace and driving safety.	Canadian Automobile Association's sentiment is Encouraging and Positive, caused by its commitment to road safety and public welfare through initiatives like the "Promise to Focus on the Road" campaign. Workplace Safety North's sentiment is Supportive and Professional, caused by its alignment with broader safety goals and its role in promoting responsible behavior among workers, particularly those who drive as part of their jobs.	1. "Promise" and "focus on the road" - The phrase "Promise to focus on the road" suggests a positive sentiment towards road safety, as it emphasizes the importance of focused driving. 2. "Ontario" and "CAA" - The association with "Ontario" and "CAA" (Canadian Automobile Association) suggests a positive sentiment towards road safety and vehicle safety, as CAA is known for promoting safe driving practices.	***Promise, #Road Safety** - **Positive Sentiment** - **Explanation:** The banners and signs are dedicated to making a promise to focus on the road, highlighting a proactive stance toward preventing accidents and promoting safe driving practices. **[CAA, Toronto]** - **Neutral to Positive Sentiment** - **Explanation:** Mentioning the CAA here suggests support or positive involvement in organizing or partnering for safety events, focusing on community and mutual well-being, which is generally seen as positive.

Figure 1: Case 1 result by MABSR-LLM and a competitor.

second-best performing model, InternVL3-8B, and the third-best, Qwen2.5-VL-7B, as shown in Figure 1, Figure 2, Figure 3, and Figure 4.

Figure 1 and Figure 2 illustrate the predicted aspect-sentiment pairs and the two-level sentiment explanations for Case 1 and Case 2. None of the aspects are explicitly mentioned in the text; instead, they must be inferred by integrating both textual and visual cues. The competitors fail to predict such open-vocabulary aspects, as they are limited to extracting words directly present in the text. In contrast, our model effectively identifies all the aspects by leveraging joint textual-visual semantics. Furthermore, in sentiment prediction, the competitors can only generate basic sentiment categories such as positive or neutral, whereas our model successfully captures all expressive sentiment labels. Regarding the two-level sentiment explanations generation, other models tend to produce generic or shallow explanations, lacking deeper insight into the underlying causes. In comparison, our model generates detailed and contextual explanations, accurately reflecting both surface-level sentiment cues and deeper social-cultural or event-specific mo-

tivations, enabled by its ability to align multimodal context with aspect sentiment reasoning.

Figure 3 and Figure 4 present the results of the reasoning paths generation task for Case 1 and Case 2. The competitors fail to capture a complete sentiment reasoning trajectory. Although Qwen2.5-VL-7B manages to produce some reasoning steps, these are largely superficial and lack the depth required to form a coherent inferential chain. In contrast, our model successfully generates more comprehensive and structured reasoning paths that closely mirror the underlying sentiment logic. This demonstrates our model’s capacity for sentiment reasoning and its ability to understand both how sentiment propagates and the contextual factors driving them. The quality of the reasoning paths further highlights our model’s alignment with human-like inference, offering interpretable and causally grounded sentiment explanations.

		Image	Text		
Input			Watch em fly #Canada #TWolves		
		Ground Truth	MABSR-LLM (Ours)	InternVL3-8B	Qwen2.5-VL-7B
Aspect-Sentiment		[('Canada', ['Proud', 'Exciting', 'Intense']), ('Andrew Wiggins', ['Focused', 'Ambitious','Confident']), ('Venezuela', ['Competitive', 'Respected'])]	[('Canada', ['Proud', 'Exciting', 'Intense']), ('Andrew Wiggins', ['Focused', 'Ambitious','Confident']), ('Venezuela', ['Competitive', 'Respected'])]	(Canada, Positive)	("Canada", "Positive"), ("TWolves", "Neutral")
Perceptual Causes		Canada's sentiment is Proud,Exciting, and Intense, and the cause for this sentiment is the dynamic moment in a game captured with national representation and competitive energy. Andrew Wiggins' sentiment is Focused, Determined, and Confident, and the cause for this sentiment is his mid-air shot attempt under a bright spotlight, showing intense concentration and athletic commitment. Venezuela's sentiment is Challenging , Competitive, and Respected, and the cause for this sentiment is their active defense against Canada, contributing to the tension and mutual respect in the match.	Canada's sentiment is Proud,Exciting , and Intense, and the cause for this sentiment is a dynamic game moment that reflects national identity and a spirit of competition. Andrew Wiggins' sentiment is Exciting, Focused, and Determined, and the cause for this sentiment is his mid-action pose during a dynamic basketball play, combined with the vibrant lighting. Venezuela's sentiment is Challenging, Competitive, and Respected, and the cause for this sentiment stems from their active defense against Canada, which added to the match's tension and fostered mutual respect.	The image shows a basketball game where a player from the team "Canada" is performing a dunk. The text "#Canada #TWolves" suggests a positive sentiment towards both entities. The player's action of dunking the ball is likely to be seen as a positive moment in the game, and the team name "TWolves" is associated with a positive sentiment	**Canada's Positive Sentiment**: The player achieving a slam dunk is the focal point of excitement and skill, which is advantageous and positive for Canada's representation. This action might be a scoring moment or a visually impressive highlight, enhancing the perception of Canada in the tournament. - **TWolves Positive Sentiment**: By highlighting "TWolves," it suggests a comparison or alignment with skilled players typically associated with the Minnesota Timberwolves. This implies high performance and potentially drawing parallels to recognized talent, adding to a positive narrative.
	Cognitive Causes	Canada's sentiment is Proud, Exciting , and Determined, the cause for this sentiment is the skilled performance of players and Canada's increasing prominence in international competitions. Andrew Wiggins' sentiment is Inspirational, Ambitious, and the cause for this sentiment is his success as a Canadian NBA player and his role in elevating the profile of Canadian basketball. Venezuela's sentiment is Respected, Intense and the cause for this sentiment is the competitive nature of their matches against Canada and the mutual admiration between the teams.	Canada's sentiment is Proud, Exciting , and Determined, the cause for this sentiment is he players' impressive performance and Canada's rising status on the international stage. Andrew Wiggins' sentiment is Inspirational, Ambitious, and the cause for this sentiment is his achievements in the NBA and his impact on enhancing the visibility of Canadian basketball. Venezuela's sentiment is Respected, Intense and the cause for this sentiment is the competitive nature of their matches against Canada and the mutual admiration shared between the teams.	The image shows a basketball game in progress. A player from the team "Canada" is performing a slam dunk. The player is wearing a jersey with the number 8 and the name "Bell" on it. The opposing team is "Venezuelan," as indicated by the players' jerseys.	1. **Canada:** Positive- **Reasoning:** The sentiment is positive because mentioning "Canada" alongside a hashtag related to toplevel performance implies admiration and support for the Canadian team. 2. **TWolves (likely a reference to the Minnesota Timberwolves): **Reasoning:** There is no direct positive or negative sentiment explicitly tied to the "TWolves." If the context is about comparing Canadian players to players from the Timberwolves, there may be an implied neutral or slightly negative sentiment, as it could suggest ambition or potential for improvement. However, without direct context, it leans more neutral.

Figure 2: Case 2 result by MABSR-LLM and a competitor.

Reasoning Paths	
Ground Truth	<think>Let's solve this problem step by step. Firstly, we need to obtain the historical background information related to the image and sentence: In 2015, during the Canadian National Exhibition (CNE) at Exhibition Place in Toronto, a road safety campaign was promoted to raise awareness about safe driving. The CNE, first held in 1879, is a major annual event featuring entertainment, food, and exhibitions, drawing large crowds and increasing traffic. The campaign, supported by organizations such as the Canadian Automobile Association (CAA) and Workplace Safety North, emphasized the dangers of driver distraction. Participants in the campaign posed with signs promoting focused driving, using hashtags like #focusontheroad and #Drivesafe. The Angel of Victory monument served as the backdrop for the event. These efforts aimed to reduce accidents in high-traffic areas during the busy fair season.. Secondly, we extract from the historical background information the relevant aspects: ['Canadian automobile association', 'Workplace safety north']. Thirdly, we obtain the surface information related to the image and sentence, that is, the intuitive visual experience of the image including the aesthetic information of the image: The lighting of the image is very warm and inviting, casting a soft glow that highlights the expressions of joy and engagement on the faces of the individuals. The bright, vibrant colors in the background, such as the deep blue of the backdrop and the colorful signs they are holding, create a lively and energetic atmosphere. The overall composition is well-lit, ensuring that the details of their clothing, accessories, and the props they are holding are clearly visible, contributing to the cheerful and positive tone of the photograph., the facial expressions of the characters, the scene information of the image, and the intuitive meaning of the text. Next, we identify each aspect's sentiment and the intuitive causes behind it: 1. Canadian Automobile Association's sentiment is positive, supportive, and responsible, and the cause for this sentiment is its promotion of road safety through the campaign "Promise to Focus on the Road," which aims to reduce driving distractions and foster a culture of safe driving. 2. Workplace Safety North's sentiment is professional, dedicated, and committed, and the cause for this sentiment is its involvement in educating drivers and employers about workplace-related road safety, contributing to an engaging approach to raising awareness. Finally, we utilize the historical background information from the previous step to analyze each aspect's sentiment and their deeper perspective causes: 1. Canadian Automobile Association's sentiment is neutral/professional, and the cause for this sentiment is its commitment to public welfare and safety through promoting road safety and educating drivers about safe practices. 2. Workplace Safety North's sentiment is neutral/professional, and the cause for this sentiment is its professional dedication to reducing accidents and promoting safe behaviors as part of its duty and responsibility in workplace and driving safety. Therefore, the final aspects and sentiment that can be obtained are: </think>[('Canadian Automobile Association', ['Supportive', 'Encouraging']), ('Workplace Safety North', ['Supportive', 'Uplifting'])]
MABSR-LLM (Ours)	<think>Let's solve this problem step by step. Firstly, we need to obtain the historical background information related to the image and sentence: The Canadian Automobile Association (CAA), founded in 1906, is a major advocacy group for drivers in Canada, promoting road safety and responsible driving. In 2015, the CAA partnered with the Canadian National Exhibition (CNE) in Toronto, one of North America's largest fairs, to promote safe driving during the event. The CNE, held annually in August, attracts millions of visitors and includes a large automotive section. The campaign, supported by the Workplace Safety and Insurance Board (WSIB), used the hashtag #Promise to encourage attendees to avoid distracted driving. The image shows individuals holding signs promoting the message "I promise to not drive distracted," reflecting the CAA's ongoing efforts to raise awareness about road safety. This initiative aligns with broader national campaigns aimed at reducing accidents caused by distractions such as texting or using mobile devices while driving.. Secondly, we extract from the historical background information the relevant aspects: ['Canadian automobile association', 'Workplace safety north']. Thirdly, we obtain the surface information related to the image and sentence, that is, the intuitive visual experience of the image including the aesthetic information of the image: The lighting of the image is very warm and inviting, casting a soft glow that highlights the subjects and the background elements. The vibrant blue backdrop contrasts beautifully with the warm tones of the foreground, creating a balanced and visually appealing composition. The lighting accentuates the expressions on the faces of the individuals, emphasizing their smiles and engagement, which adds to the overall positive and uplifting mood of the photograph. The interplay of light and shadow enhances the details of the signs and the architectural elements in the background, making the scene feel dynamic and engaging., the facial expressions of the characters, the scene information of the image, and the intuitive meaning of the text. Next, we identify each aspect's sentiment and the intuitive causes behind it: 1. Canadian Automobile Association's sentiment is positive, supportive, and advocative, and the cause for this sentiment is its active promotion of road safety through initiatives like the "Promise to not drive distracted" campaign, as well as its visible presence at the CNE with a professional and approachable image. 2. Workplace Safety and Insurance Board's sentiment is professional, authoritative, and caring, and the cause for this sentiment is its endorsement of the road safety campaign, reflecting its commitment to protecting workers and the public by addressing distracted driving as a serious occupational hazard. Finally, we utilize the historical background information from the previous step to analyze each aspect's sentiment and their deeper perspective causes: 1. Canadian Automobile Association's sentiment is positive, and the cause for this sentiment is its commitment to promoting road safety and responsible driving through initiatives like the "Promise to not drive distracted" campaign. 2. Workplace Safety and Insurance Board's sentiment is neutral to positive, and the cause for this sentiment is its collaboration with the CAA in the campaign, reflecting a shared commitment to public safety and reducing workplace-related accidents caused by distracted driving. Therefore, the final aspects and sentiment that can be obtained are: </think>[('Canadian Automobile Association', ['Supportive', 'Encouraging']), ('Workplace Safety North', ['Supportive', 'Uplifting'])]
InternVL3-8B	1. Image Analysis: - The image shows a group of people holding signs that say "I Promise to Focus on the Road" and "My Promise, Focus on the Road." - The background features a large structure resembling a monument, with a blue sky and flags. - The image is associated with a campaign called "Promise to Focus on the Road," with hashtags #Promise #focusontheroad and #CNE2015. 2. Entities and Sentiments: - Entities: - CAA (Canadian Automobile Association) - Government of Ontario - Government of Canada - Sentiments: - Promoting road safety and awareness. - Encouraging responsible driving. - Supporting a safe and responsible driving culture. 3. Output: - The image is promoting road safety and awareness. - The campaign is supported by various organizations and government bodies in Ontario and Canada. This analysis considers the visual content, the message conveyed by the signs, and the associated hashtags to determine the entities and sentiments.
Qwen2.5-VL-7B	Step 1: Identify Entities - Parents- Child- Driver- CAA- Ontario- Workplace Safety & Insurance Board- IHSA- #LetsGoToTheEX- Toronto- 2015- CNE2015- Drivesafe Step 2: Determine Sentiment for Each Entity 1. Parents: - Positive sentiment; parents are supportive. 2. Child: - Positive sentiment; child is enthusiastic. 3. Driver: - Neutral to slightly positive; the message is aimed at focusing. 3. CAA (Canadian Automobile Association): - Neutral; sponsor/partner. 4. Ontario: - Neutral; location context. 5. #LetsGoToTheEX: - Neutral; a hashtag indicating attendance at an event. 6. Toronto: - Neutral; location as a city context. 7. 2015: - Neutral; year as a year-specific context. 8. CNE2015 (Canadian National Exhibition 2015): - Positive; related to a popular event.

Figure 3: Case 1 reasoning paths result by MABSR-LLM and a competitor.

Reasoning Paths

Ground Truth	<think>Let's solve this problem step by step. Firstly, we need to obtain the historical background information related to the image and sentence: The image shows a Canadian basketball player mid-dunk during an international game, with the FIBA logo visible and Venezuela as the opposing team. FIBA, the governing body for international basketball, organizes major events like the World Cup and Olympic competitions. Canada has increasingly participated in these tournaments since the 2000s, with notable players such as Andrew Wiggins and Jamal Murray representing the nation. The hashtag #TWolves suggests a connection to the Minnesota Timberwolves, while the Toronto Raptors have also contributed to Canadian basketball development. Venezuela is a frequent rival in FIBA Americas competitions, where teams compete for qualification to global events. The scene reflects the growing competitiveness of Canadian basketball on the international stage.. Secondly, we extract from the historical background information the relevant aspects: ['Canada', 'Andrew wiggins', 'Venezuela']. Thirdly, we obtain the surface information related to the image and sentence, that is, the intuitive visual experience of the image including the aesthetic information of the image: The lighting of the image is very vibrant and dynamic, casting a warm glow that highlights the intensity of the moment. The bright spotlights focus on the player in mid-air, emphasizing his athletic form and the energy of the game. The contrast between the illuminated court and the darker background creates a dramatic effect, drawing attention to the action at the basket. The colors are vivid, with the reds and whites of the uniforms standing out sharply against the neutral tones of the arena, adding to the visual appeal and excitement of the scene. Overall, the lighting enhances the sense of motion and passion, making the image feel alive and engaging., the facial expressions of the characters, the scene information of the image, and the intuitive meaning of the text. Next, we identify each aspect's sentiment and the intuitive causes behind it: 1. Canada's sentiment is excitement, pride, and intensity, and the cause for this sentiment is the dynamic moment in a game captured with national representation and competitive energy. 2. Andrew Wiggins' sentiment is focus, determination, and confidence, and the cause for this sentiment is his mid-air shot attempt under a bright spotlight, showing intense concentration and athletic commitment. 3. Venezuela's sentiment is challenge, competitiveness, and respect, and the cause for this sentiment is their active defense against Canada, contributing to the tension and mutual respect in the match. Finally, we utilize the historical background information from the previous step to analyze each aspect's sentiment and their deeper perspective causes: - Canada's sentiment is pride, excitement, and determination, and the cause for this sentiment is the skilled performance of players and Canada's increasing prominence in international competitions. - Andrew Wiggins' sentiment is inspiration, ambition, and representation, and the cause for this sentiment is his success as a Canadian NBA player and his role in elevating the profile of Canadian basketball. - Venezuela's sentiment is respect, intensity, and rivalry, and the cause for this sentiment is the competitive nature of their matches against Canada and the mutual admiration between the teams. Therefore, the final aspects and sentiment that can be obtained are: </think>[('Canada', ['Proud', 'Exciting', 'Intense']), ('Andrew Wiggins', ['Focused', 'Ambitious', 'Confident']), ('Venezuela', ['Competitive', 'Respected'])]
MABSR-LLM (Ours)	<think>Let's solve this problem step by step. Firstly, we need to obtain the historical background information related to the image and sentence: The image captures a basketball game at the 2015 FIBA World Cup, held in Spain from August 30 to September 14. The Canadian team, wearing blue jerseys with "Canada" and "Bell" branding, is seen in action against another team, likely Venezuela, as indicated by the "Venezuela" logo on the scoreboard. The game took place at the Palau Sant Jordi in Barcelona, one of the tournament's main venues. Canada, led by head coach Kenny Atkinson, reached the semifinals for the first time since 1994, marking a significant achievement. Key players included Cory Joseph, Kyle Nurse, and Chris Boucher. The Bell logo reflects the sponsorship role of Bell Canada, a major telecommunications company supporting the team during the event.. Secondly, we extract from the historical background information the relevant aspects: ['Canada', 'Andrew wiggins', 'Venezuela']. Thirdly, we obtain the surface information related to the image and sentence, that is, the intuitive visual experience of the image including the aesthetic information of the image: The lighting of the image is very vibrant and dynamic, casting a warm glow that highlights the intensity of the basketball game. The bright stadium lights illuminate the players on the court, creating sharp contrasts between the illuminated areas and the darker shadows, which adds depth and energy to the scene. The colors are vivid, with the blue jerseys of the Canadian team standing out against the red and white of the opposing team, enhancing the visual appeal. The overall atmosphere is lively and engaging, capturing the excitement of the moment as the player dunks the ball, showcasing the athleticism and passion of the sport., the facial expressions of the characters, the scene information of the image, and the intuitive meaning of the text. Next, we identify each aspect's sentiment and the intuitive causes behind it: Canada's sentiment is characterized by excitement, pride, and intensity, stemming from a dynamic in-game moment that highlights national representation and competitive energy. Andrew Wiggins' sentiment reflects focus, determination, and confidence, driven by his mid-air shot attempt under a bright spotlight, which showcases his intense concentration and athletic dedication. Venezuela's sentiment conveys challenge, competitiveness, and respect, arising from their active defense against Canada, which contributed to the match's tension and the mutual respect between the teams. Finally, we utilize the historical background information from the previous step to analyze each aspect's sentiment and their deeper perspective causes: Canada's sentiment reflects pride, excitement, and determination, stemming from the players' skilled performances and the nation's rising prominence in international competitions. Andrew Wiggins' sentiment conveys inspiration, ambition, and representation, driven by his success as a Canadian NBA player and his role in raising the profile of Canadian basketball. Venezuela's sentiment embodies respect, intensity, and rivalry, resulting from the competitive nature of their matches against Canada and the mutual admiration shared between the teams. Therefore, the final aspects and sentiment that can be obtained are: </think>[('Canada', ['Proud', 'Exciting', 'Intense']), ('Andrew Wiggins', ['Focused', 'Ambitious', 'Confident']), ('Venezuela', ['Competitive', 'Respected'])]
InternVL3-8B	1. Image Analysis: - The image shows a basketball game in progress. - A player from the team "Canada" is performing a slam dunk. - The player is wearing a jersey with the number 8 and the name "Bell." - The opposing team is "Venezuelan," as indicated by their jerseys. 2. Sentiment Analysis: - The action of the player slam dunking suggests a positive sentiment. - The presence of the word "Fly" in the hashtag implies a sense of excitement or achievement. - The mention of "Canada" and "TWolves" indicates a connection to the team and possibly a sense of pride or support. 3. Entities and Sentiments: - Entities: - Player (Canada) - Team (Canada) - Opposing Team (Venezuelan) - Event (Basketball Game) - Sentiments: - Positive (Slam dunk, excitement) - Neutral (Game context)
Qwen2.5-VL-7B	### Observations from the Image: 1. Basketball Scene: The image shows a basketball game in progress. 2. Player from Canada: The player wearing a white jersey with "Canada" and the number "8" is performing a slam dunk. 3. Opponent: The opposing team, likely Venezuela, is in red and maroon jerseys. ### Analysis of the Sentence: 1. Watch em fly: - Sentiment: Encouraging and enthusiastic. - Cause: Highlights the impressive athletic ability of the Canadian player seen dunking the ball. 2. Canada: - Sentiment: Supportive. - Cause: Indicates national pride and support for the Canadian team in the event. 3. TWolves: - Sentiment: Could be neutral or curious, depending on the context. - Cause: Refers to the Minnesota Timberwolves, a professional basketball team. The term could bring context to where the player is from, but it might also be referring to a player's style or being associated with the Timberwolves team indirectly. ### List of Entities and Sentiments: 1. Canada: - Sentiments: Positive (national pride). TWolves (Minnesota Timberwolves): - Sentiments: Neutral/Inquisitive (contextual reference)

Figure 4: Case 2 reasoning paths result by MABSR-LLM and a competitor.

F Ethic Considerations

In developing the OX-MABSR benchmark and modeling framework, we ensured ethical data handling and responsible research conduct. All data were sourced from public platforms or obtained with informed consent, with contributors made aware of their rights, including data withdrawal. Content use complied with platform terms and licensing policies, and official APIs were used where applicable. To promote open and inclusive research, all datasets, models, and code will be released under open-source licenses. We also acknowledge the potential for social or cultural biases in multimodal sentiment data and models. To mitigate this, we conducted careful data curation and model evaluation across diverse scenarios and demographic contexts. We encourage further analysis and community feedback to ensure fairness and reduce unintended bias in future iterations. Fairness and inclusivity are ongoing commitments that require continuous attention. We welcome collaboration to audit and improve our benchmark and models over time.

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